

Research Experience

- 📌 **Cold Atoms Intern**, Laboratoire Photonique Numérique et Nanosciences (LP2N)
05/2024 – 07/2024
Supervisor: Andrea Bertoldi
 - Locking of a 780 nm laser diode to a ^{87}Rb D2 transition for the Cryst³ experiment
 - Saturated absorption and frequency modulation spectroscopy
 - Frequency locking of the laser with an analog PID control loop
- 📌 **Theory Research Intern**, Laboratoire Ondes et Matière d'Aquitaine (LOMA)
06/2023 – 07/2023
Supervisors: Fabio Pistolesi & Rémi Avriller
 - Studied perturbative methods for solving non-linear oscillator systems
 - Written report: Synchronisation of two weakly coupled Van der Pol oscillators

Education

- 2024 – 📌 **Erasmus BSc exchange**, Lund University
- 2022 – 📌 **BSc, Physics**, Université de Bordeaux
First year average: **171/200** | Rank: **3/81**
Second year average: **173/200** | Rank: **1/10**
- 2020 – 2022 📌 **IB Diploma**, Dulwich College Shanghai Pudong.
Final grade: **42/45**

Languages

- English 📌 Bilingual (IELTS Academic - C2)
- French 📌 Bilingual
- Mandarin Chinese 📌 Conversational

Programming Experience

- Python 📌 **Scientific programming** 🐙 [Link](#)
This repository is a collection of python scripts and jupyter notebooks I have created for university classes, labs, data visualisation and some simulations. It is classified by year (L1, L2, L3) then by semester (S1 - S5).
- Julia 📌 **Multi-layered neural networks** 🐙 [Link](#)
From scratch neural network implementations. Notably with the programming of a perceptron used for classifying images and a deep neural net with backpropagation.
- C++ 📌 **OpenGL Graphics programming** 🐙 [Link](#)
Low-level rendering engine. Textures and Cameras implemented. Inspiration taken from Hazel engine development vlogs.

Programming Experience (continued)

Java  **Robotics Programming (First Tech Challenge)**  [Link](#)

Team captain and programmer for highschool robotics team. Implementation of autonomous phases and PID motor control.

Other Experiences

10/24 –  **Student Pub Foreman**, Kalmar student nation (Lund),

- Management of the bar and kitchen
- Teaching new pub volunteers

05/24 – 07/24  **ScienceL@b 22 Co-founder**, Université de Bordeaux

- Student-run laboratory in the A22 licence building
- Management of inventory

09/23 – 05/24  **Student Life Office Employee**, Université de Bordeaux

- Running university events, organisation of the inventory
- Orientation of new students

Miscellaneous

Awards and Achievements

2023  **Internship Scholarship for Academic Excellence**, Université de Bordeaux,

Internship scholarship awarded by the Bachelor's Department, leading to an internship report and an oral presentation. 4 week intership carried out at the Laboratoire Ondes et Matière d'Aquitaine (LOMA).

Certification

2023  **Mental Health First Aid (PSSM)**, Université de Bordeaux.

References

Available on Request